Microcontrollers

ESP32-C3

U_ESP32
ESP32.SchDoc

ESP_USB_D_N
ESP_USB_D_P
RP_ESP_GPIO
ESP_SPI_MISO
ESP_SPI_SCK
ESP_SPI_CSn
ESP_SPI_MOSI

Left Wing (1)

48 RGB LEDs + Drivers

U_16_RGBs1
16_RGBs.SchDoc
U_16_RGBs2
16_RGBs.SchDoc
U_16_RGBs3
16_RGBs.SchDoc

SPI1_CLK
LE
OE
CLK
SDI1
SDI2
SDI3

U_RP2350
RP2350.SchDoc

SPI1_0
SPI1_1
SPI1_2
SPI1_3
SPI1_4
SPI1_5
SPI1_6
SPI1_7
SPI1_8
RP_ESP32_GPIO
RP_SPI_MISO
RP_SPI_SCK
RP_SPI_CSn
RP_SPI_MOSI
HE_1
SPI1_CLK
OE
LE
RP_USB_D_P
RP_USB_D_N

Hall Effect

+5V
R9
100R
U4
VSUP
VOUT
GND
C24
4.7nF
+3V3_RP
R10
4k7
C25
1nF

Right Wing (0)

48 RGB LEDs + Drivers

U_16_RGBs4
16_RGBs.SchDoc
U_16_RGBs5
16_RGBs.SchDoc
U_16_RGBs6
16_RGBs.SchDoc

SPI0_CLK
LE
OE
SDI1
SDI2
SDI3
CLK

SPI0_0
SPI0_1
SPI0_2
SPI0_3
SPI0_4
SPI0_5
SPI0_6
SPI0_7
SPI0_8
SPI0_CLK
HE_0

Hall Effect

+3V3_RP
R11
4k7
U5
VOUT
VSUP
GND
C26
1nF
+5V
R12
100R
C27
4.7nF

RP2350

USB-C + Switching

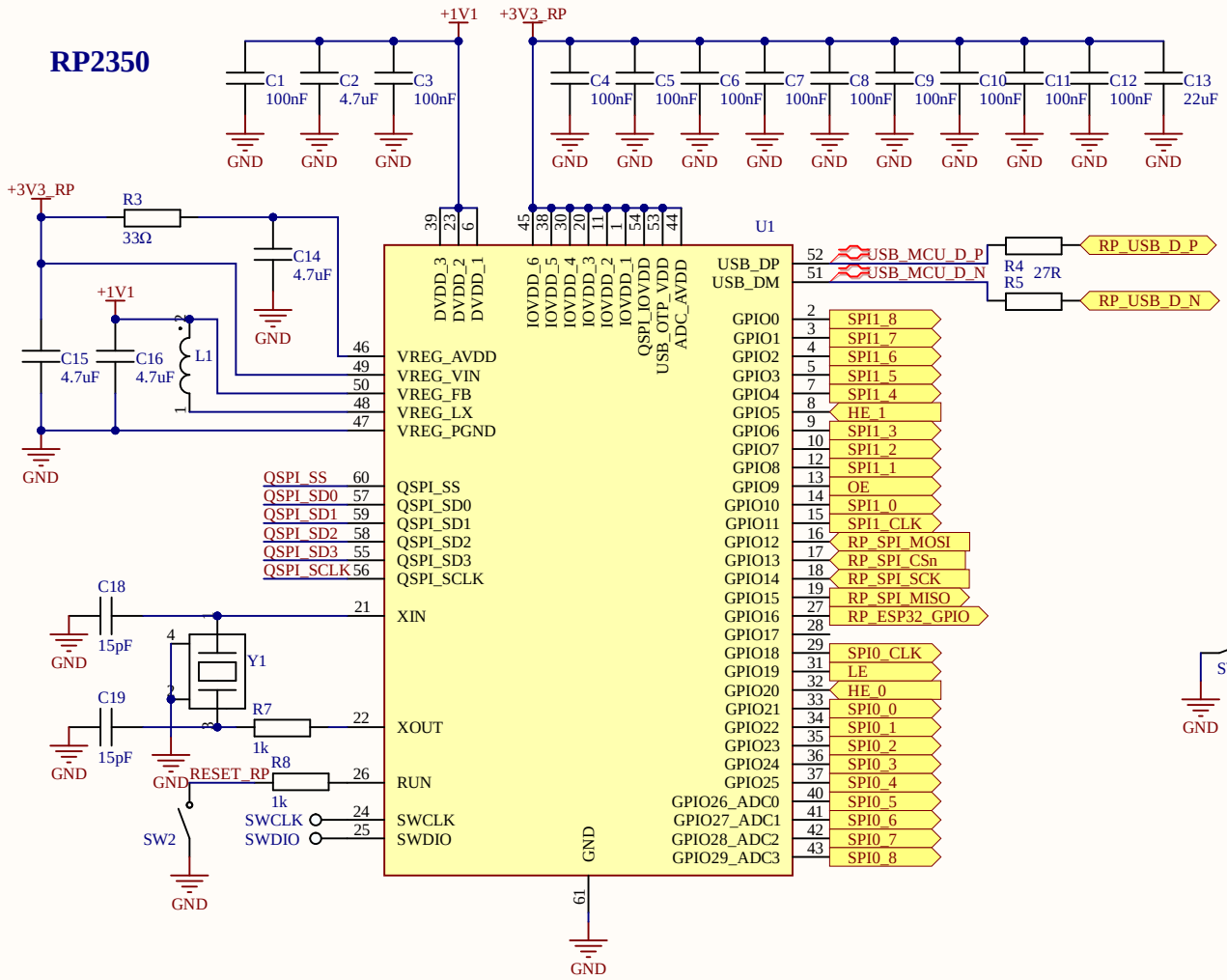
+5V
D1
R1
5.1k
R2
5.1k
USB_D_N
USB_D_P
GND
USBC1
A1B12
A4B9
B8
A5
B7
A6
A7
B6
A8
B5
B4A9
B1A12
GND
VBUS
SBU2
CC1
DN2
DP1
DN1
DP2
SBU1
CC2
VBUS
GND
EH

RP_USB_D_N
RP_USB_D_P
ESP_USB_D_N
ESP_USB_D_P
1D-
1D+
2D-
2D+
D-
D+
S
OE
VCC
U10
+3V3_RP
SW5
R16
10k
C37
100nF

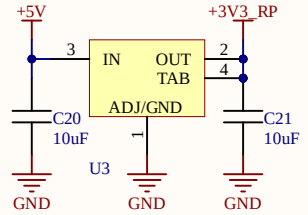
Power

+5V
C22
22uF
C23
100nF
GND_W
5V_W
D2

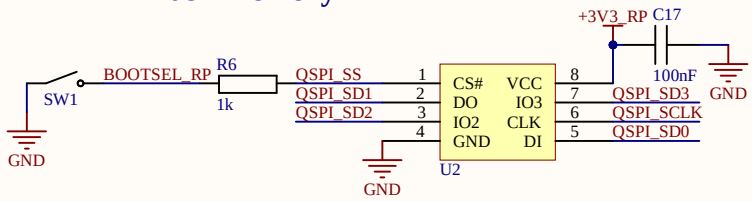
Title: POV Display - Master Sheet		
Size: A4	Number:	Revision: C
Date: 10/17/2026		Sheet 1 of 1
File: C:\Engineering\..\POV_Display.SchDoc		Drawn By: Jonathan Duncan



3.3V Regulation

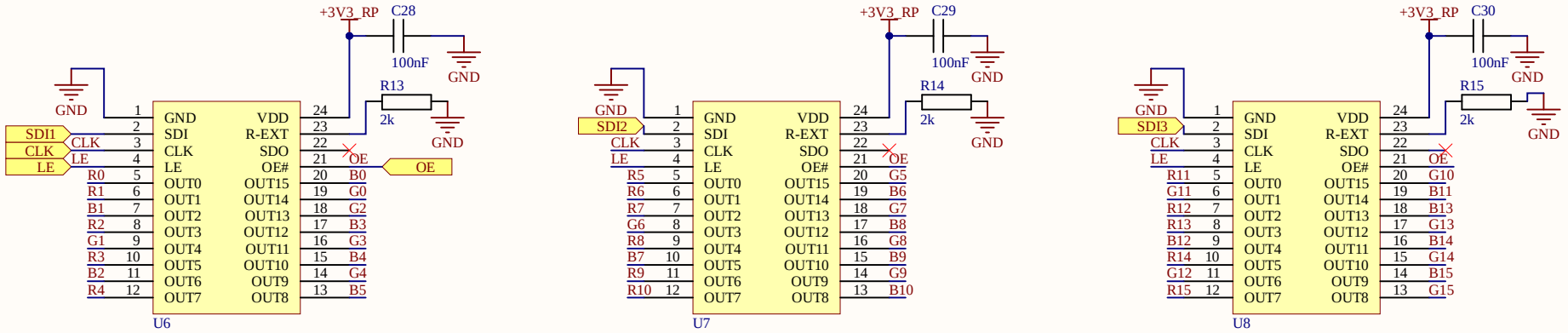


Flash Memory

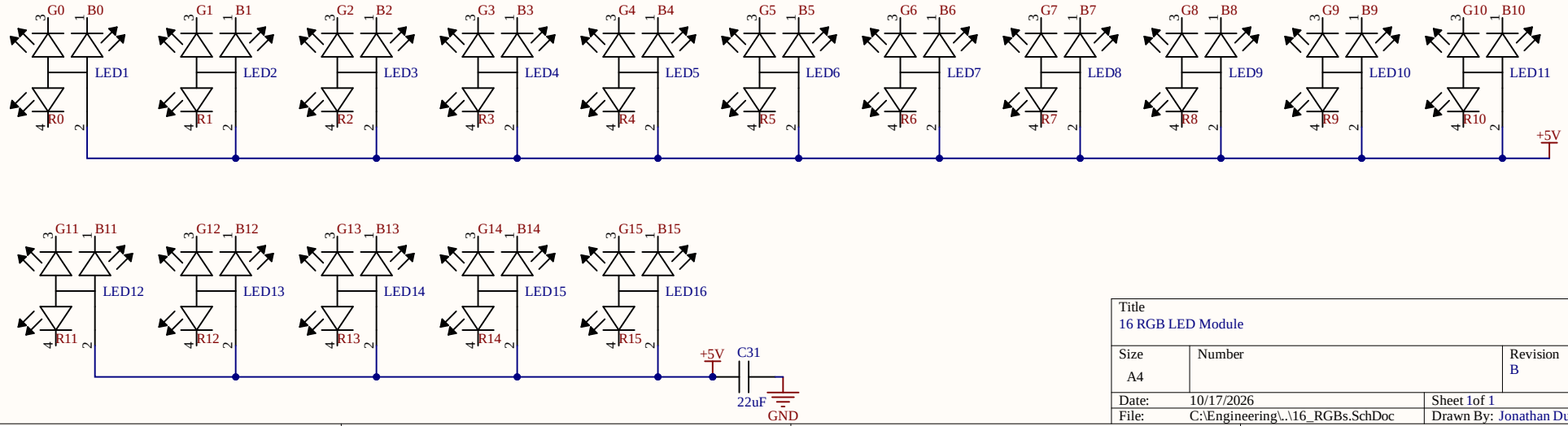


Title: RP2350 Microcontroller		
Size: A4	Number:	Revision: D
Date: 10/17/2026		Sheet 1 of 1
File: C:\Engineering\..\RP2350.SchDoc		Drawn By: Jonathan Duncan

Drivers

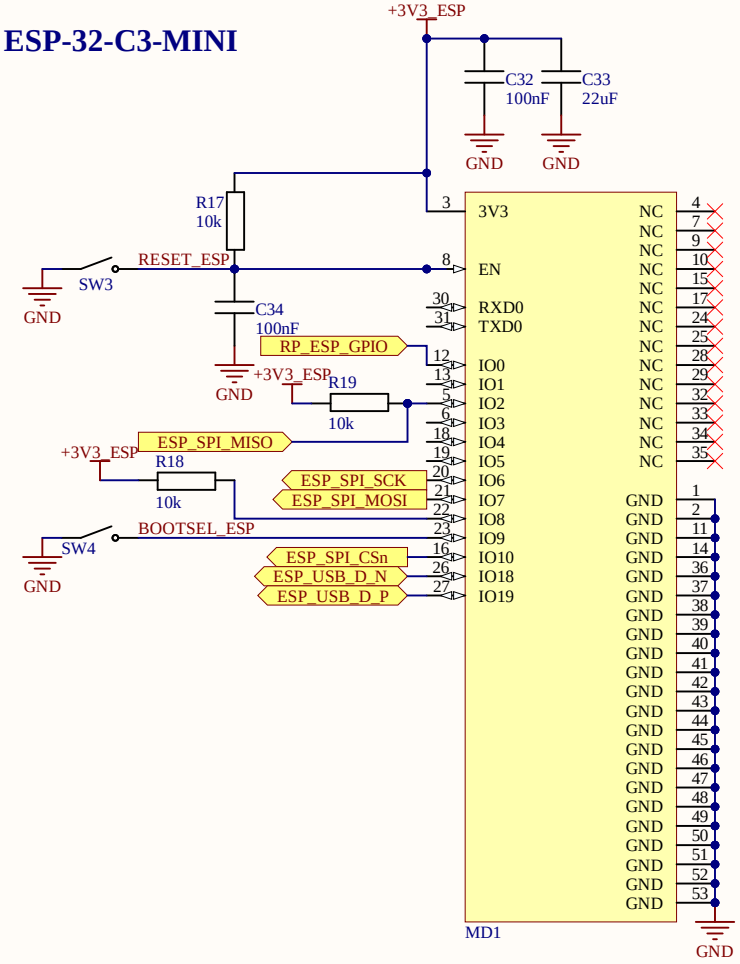


LEDs

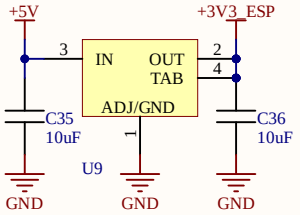


Title: 16 RGB LED Module		
Size: A4	Number:	Revision: B
Date: 10/17/2026		Sheet 1of 1
File: C:\Engineering\..\16_RGBs.SchDoc		Drawn By: Jonathan Duncan

ESP-32-C3-MINI



3.3V Regulation



Title: ESP32 Microcontroller		
Size: A4	Number:	Revision: B
Date: 10/17/2026		Sheet 1 of 1
File: C:\Engineering\..\ESP32.SchDoc		Drawn By: Jonathan Duncan